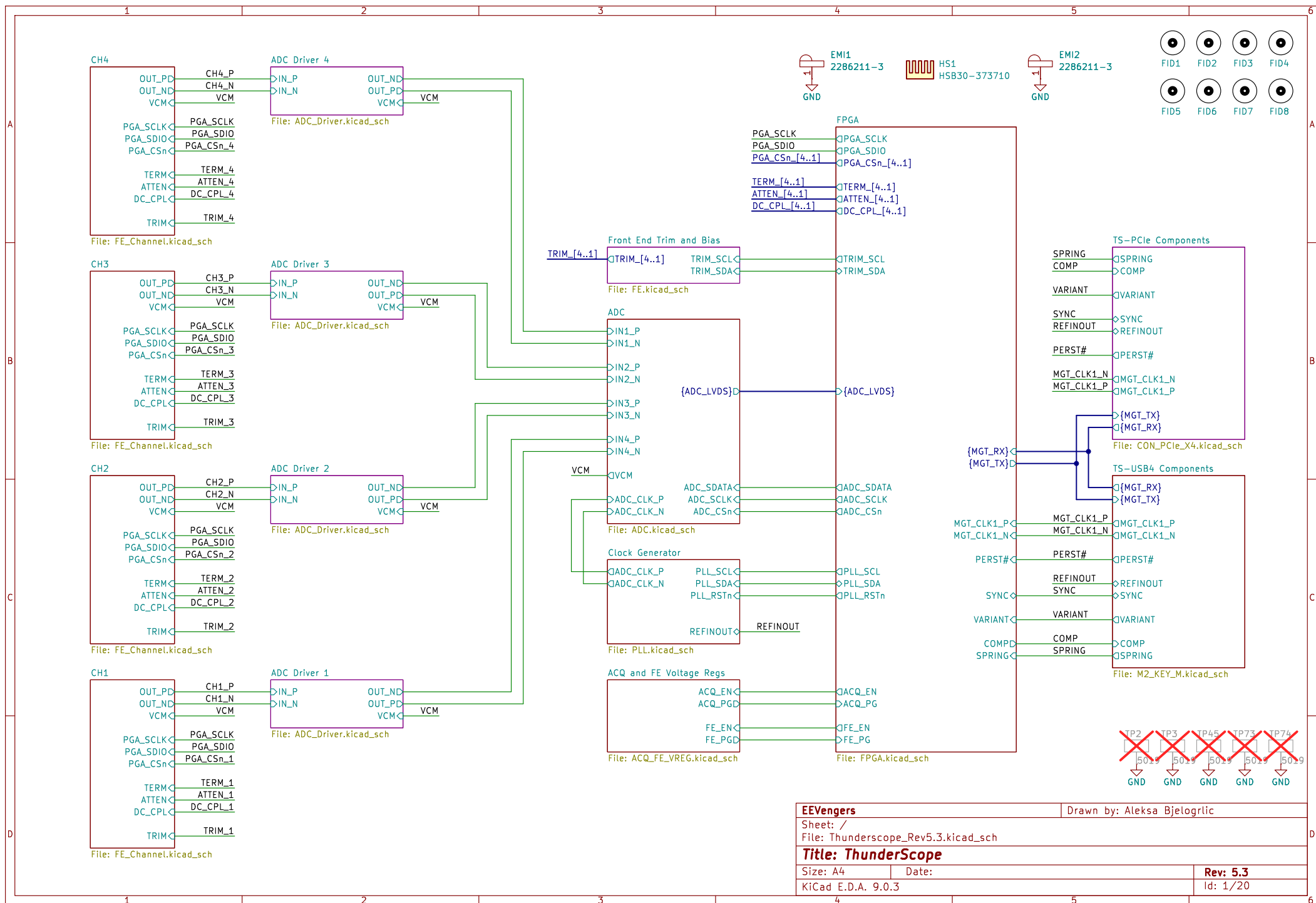




The PCB layout for the Termination and Attenuation circuit is shown. It includes a BNC input (J1002\_4) connected to a termination network (K1000\_4A) and a 50x attenuator (K1001\_4A). The termination network consists of a 30 ohm resistor (R1022\_4) and a 15 ohm resistor (R1021\_4) in series, with a 18pF capacitor (C1058\_4) and a 0 ohm resistor (R1053\_4) in parallel. The 50x attenuator consists of a 1K resistor (R1001\_4) and a 976k resistor (R1024\_4) in series, with a 3pF capacitor (C1023\_4) and a 0.3pF capacitor (C1012\_4) in parallel. The circuit also includes a USB input (+VUSB) connected to a termination network (K1000\_4C) and a 50x attenuator (K1001\_4C). The USB input network consists of a 22 ohm resistor (R1027\_4) and a 22uF capacitor (C1026\_4) in series, with a 22uF capacitor (C1027\_4) and a 1uF capacitor (C1009\_4) in parallel. The 50x attenuator network consists of a 1K resistor (R1001\_4) and a 976k resistor (R1024\_4) in series, with a 3pF capacitor (C1023\_4) and a 0.3pF capacitor (C1012\_4) in parallel. The circuit also includes a termination network (K1000\_4B) and a 50x attenuator (K1001\_4B) for the output. The output network consists of a 1K resistor (R1001\_4) and a 976k resistor (R1024\_4) in series, with a 3pF capacitor (C1023\_4) and a 0.3pF capacitor (C1012\_4) in parallel. The circuit also includes a termination network (K1000\_4C) and a 50x attenuator (K1001\_4C) for the output. The output network consists of a 1K resistor (R1001\_4) and a 976k resistor (R1024\_4) in series, with a 3pF capacitor (C1023\_4) and a 0.3pF capacitor (C1012\_4) in parallel. The circuit also includes a termination network (K1000\_4B) and a 50x attenuator (K1001\_4B) for the output. The output network consists of a 1K resistor (R1001\_4) and a 976k resistor (R1024\_4) in series, with a 3pF capacitor (C1023\_4) and a 0.3pF capacitor (C1012\_4) in parallel.

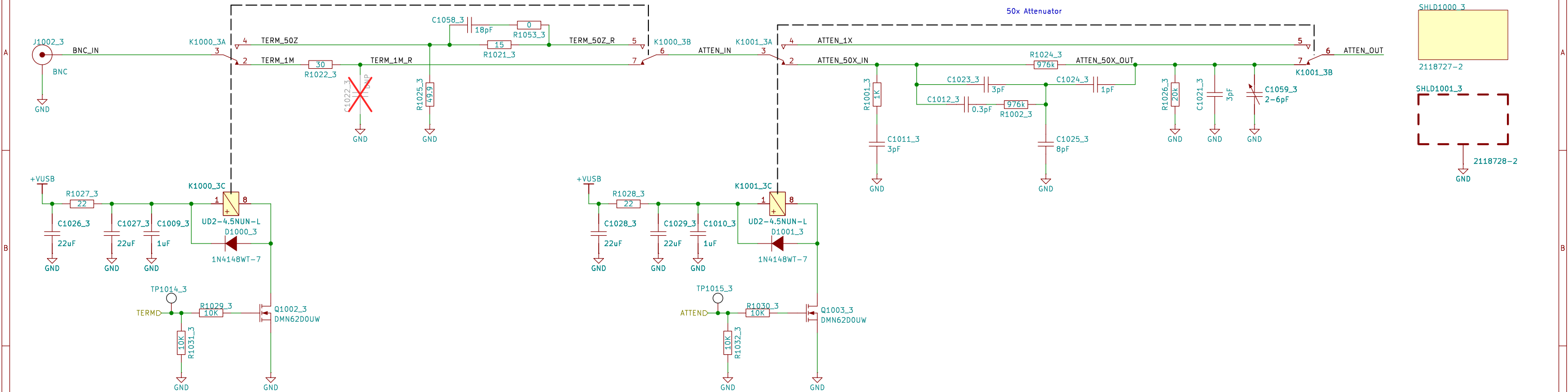
Max current for 35.7kOhm R\_Bias: 24mA @ V+ and 24mA @ V-  
100k R\_Bias will be lower: 17mA per rail was measured on the EVM

Table 8-3. TMUX4053 Truth Table

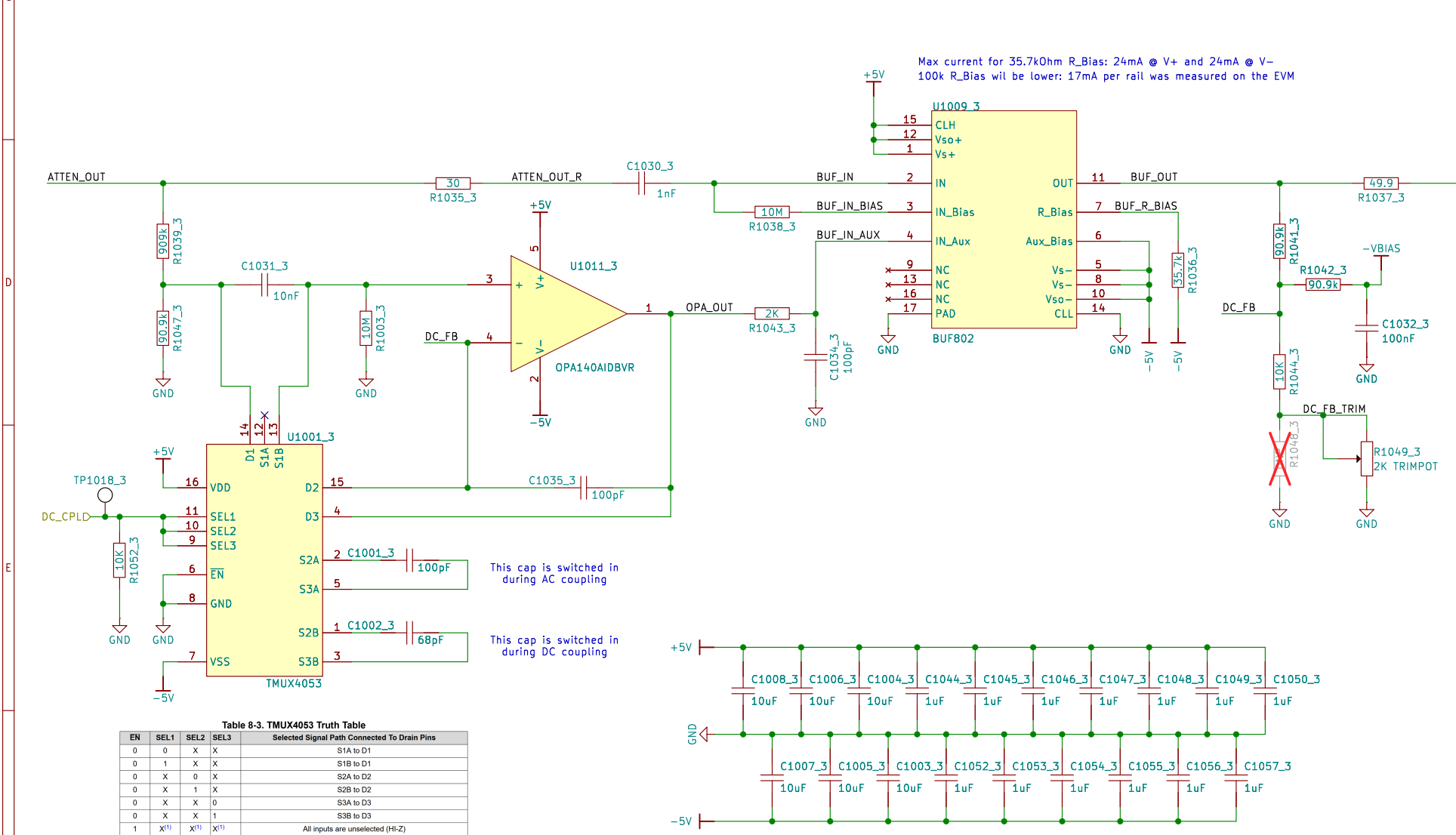
| EN | SEL1             | SEL2             | SEL3             | Selected Signal Path Connected To Drain Pins |
|----|------------------|------------------|------------------|----------------------------------------------|
| 0  | 0                | X                | X                | S1A to D1                                    |
| 0  | 1                | X                | X                | S1B to D1                                    |
| 0  | X                | 0                | X                | S2A to D2                                    |
| 0  | X                | 1                | X                | S2B to D2                                    |
| 0  | X                | X                | 0                | S3A to D3                                    |
| 0  | X                | X                | 1                | S3B to D3                                    |
| 1  | X <sup>(1)</sup> | X <sup>(1)</sup> | X <sup>(1)</sup> | All inputs are unselected (Hi-Z)             |

[illegible]

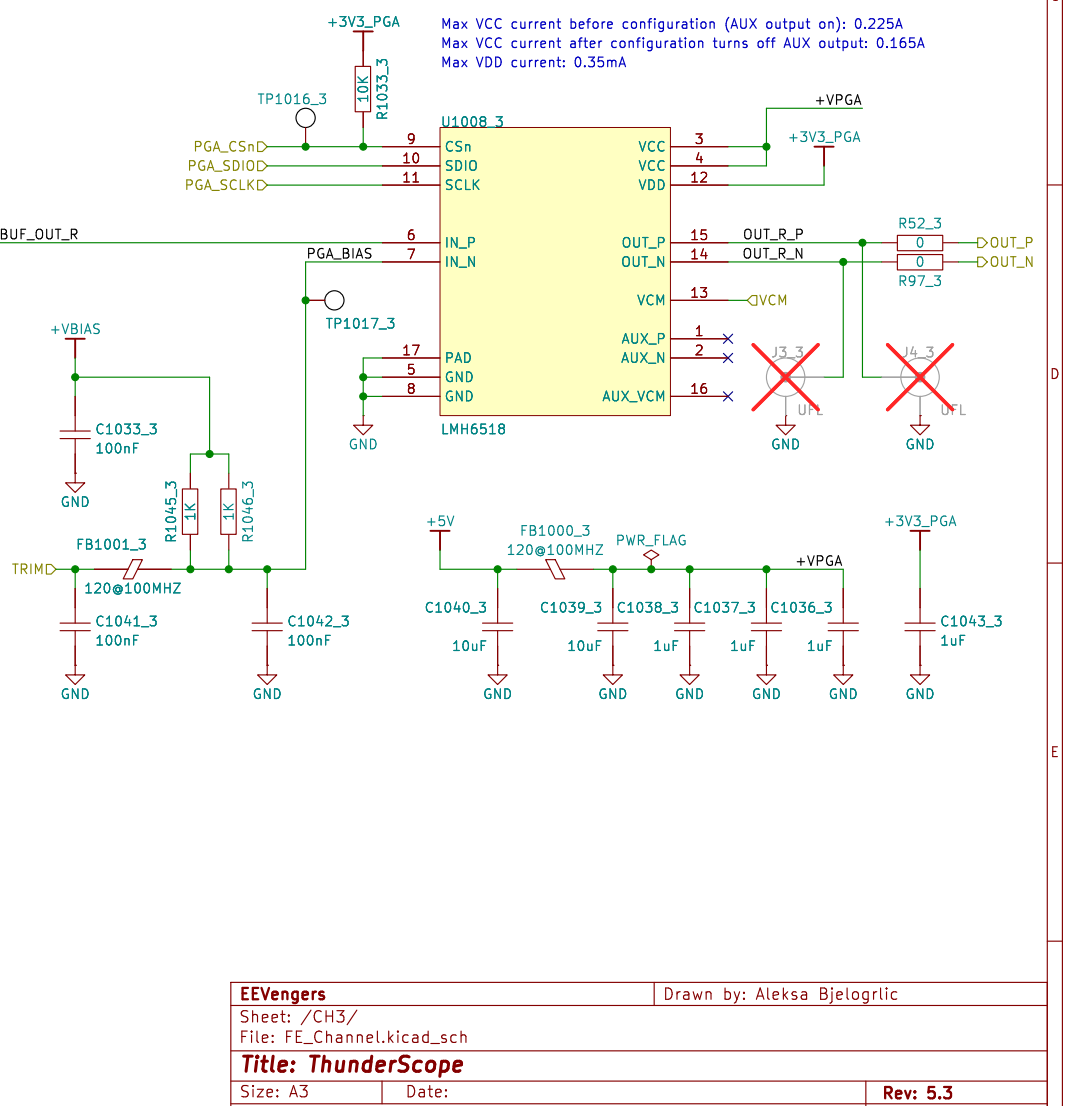
Termination and Attenuation



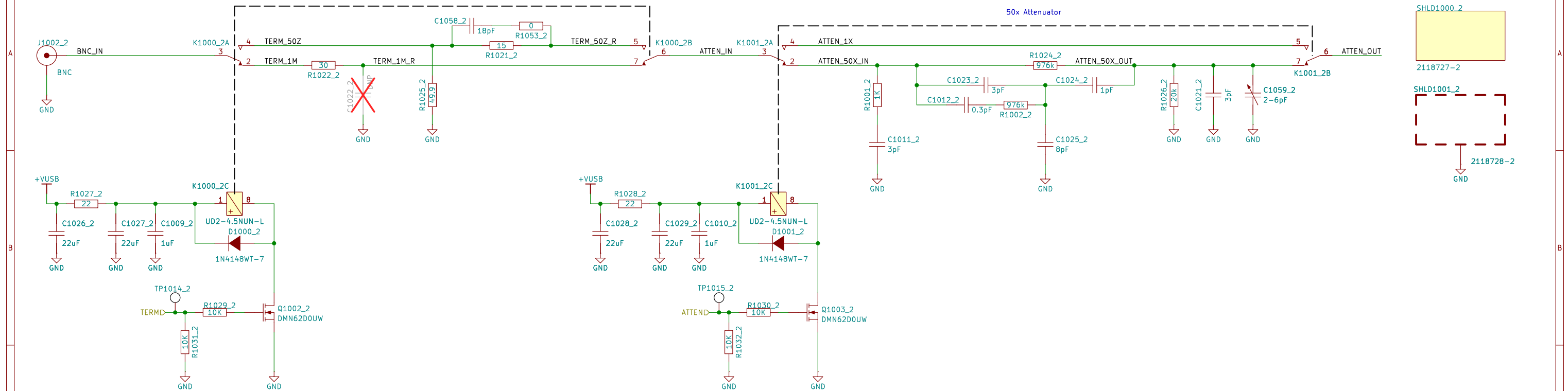
Input Buffer and AC/DC Coupling



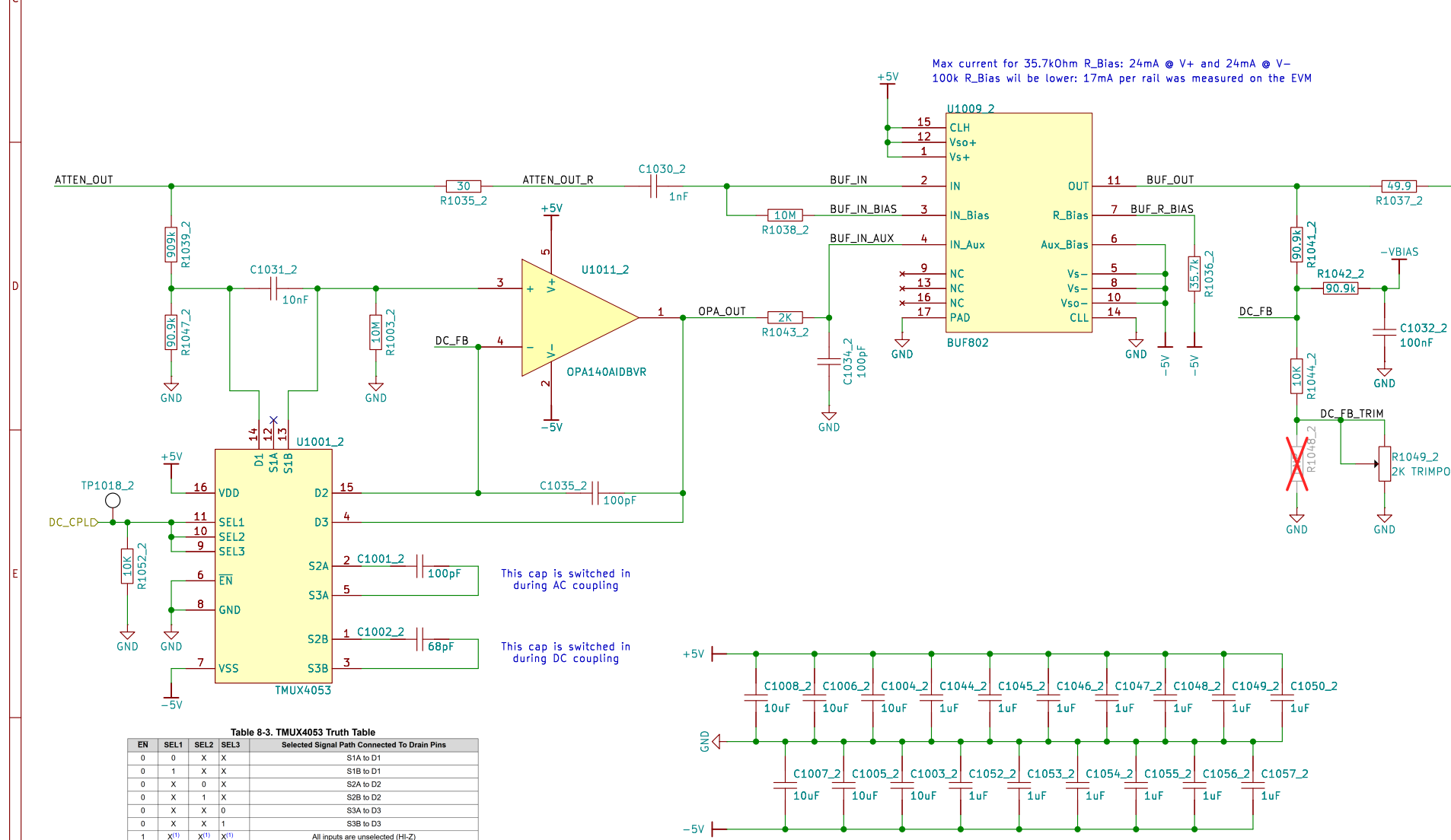
Programmable Gain Amplifier



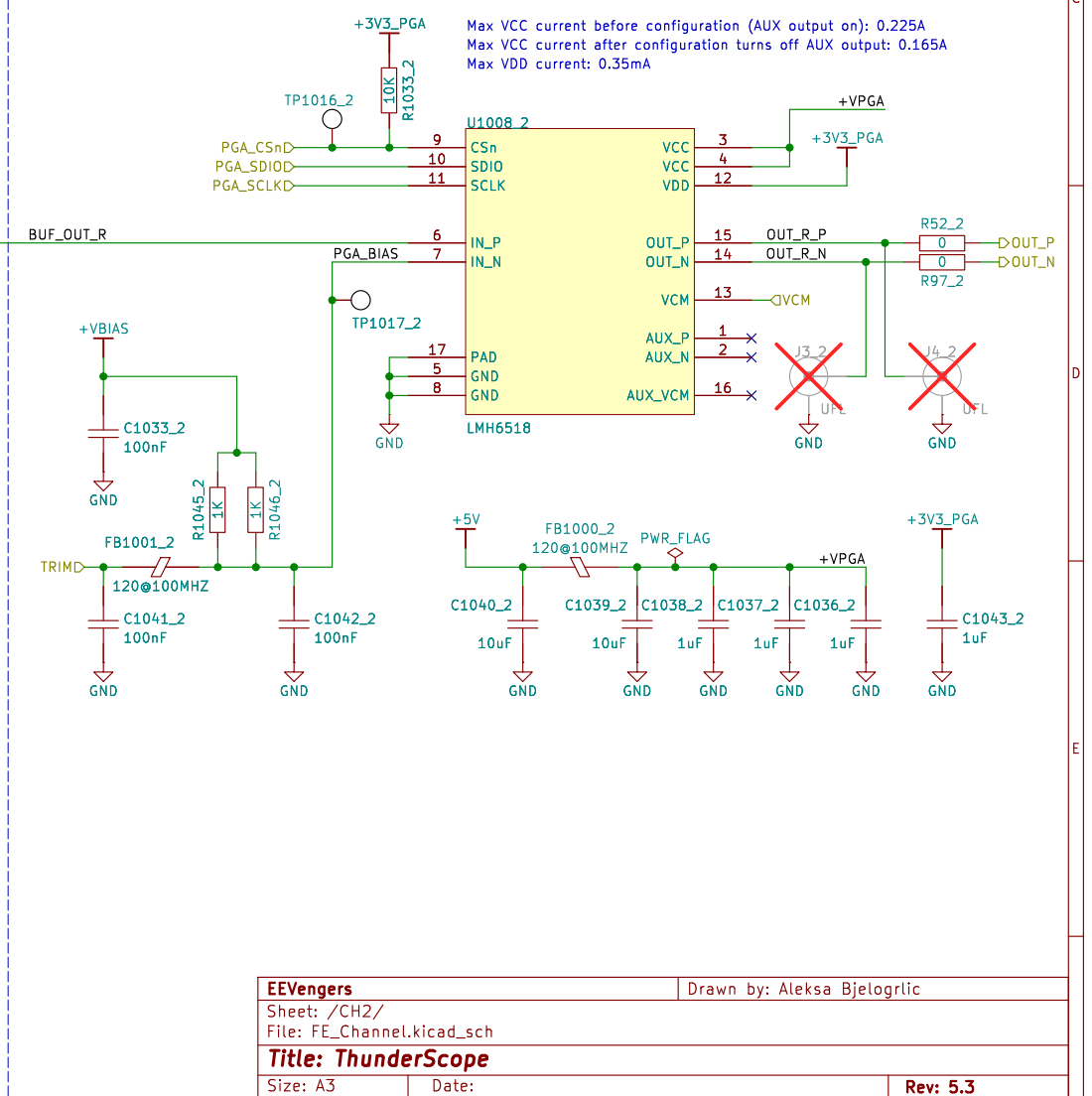
## Termination and Attenuation



## Input Buffer and AC/DC Coupling



## Programmable Gain Amplifier



The PCB layout for the Termination and Attenuation circuit is shown. It includes a BNC input (J1002\_1) connected to a termination network (K1000\_1A, K1000\_1B) and a 50x Attenuator (K1001\_1A, K1001\_1B). The circuit also features a USB power input (+VUSB) connected to a diode (UD2-4.5NUN-L D1000\_1) and a MOSFET (Q1002\_1 DMN62D0UW) for signal processing. Various passive components like resistors (R1027\_1, R1028\_1, R1029\_1, R1031\_1, R1032\_1, R1053\_1, R1021\_1, R1022\_1, R1024\_1, R1025\_1, R1026\_1, R1030\_1) and capacitors (C1026\_1, C1027\_1, C1028\_1, C1029\_1, C1030\_1, C1031\_1, C1032\_1, C1058\_1, C1059\_1, C1023\_1, C1024\_1, C1025\_1, C1021\_1) are used for signal conditioning and termination. The layout is divided into two main sections: Termination and Attenuation, with a 50x Attenuator section in the center.

Max current for 35.7kOhm R\_Bias: 24mA @ V+ and 24mA @ V-  
100k R\_Bias will be lower: 17mA per rail was measured on the EVM

Table 8-3. TMUX4053 Truth Table

| EN | SEL1             | SEL2             | SEL3             | Selected Signal Path Connected to Drain Pins |
|----|------------------|------------------|------------------|----------------------------------------------|
| 0  | 0                | X                | X                | S1A to D1                                    |
| 0  | 1                | X                | X                | S1B to D1                                    |
| 0  | X                | 0                | X                | S2A to D2                                    |
| 0  | X                | 1                | X                | S2B to D2                                    |
| 0  | X                | X                | 0                | S3A to D3                                    |
| 0  | X                | X                | 1                | S3B to D3                                    |
| 1  | X <sup>(1)</sup> | X <sup>(1)</sup> | X <sup>(1)</sup> | All inputs are unselected (Hi-Z)             |

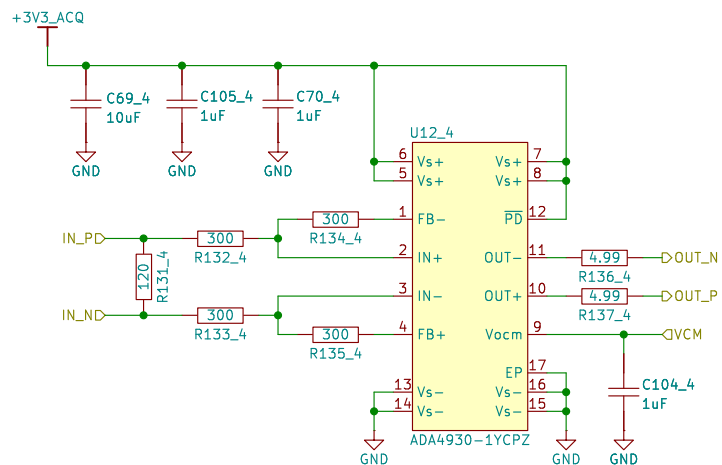
**Programmable Gain Amplifier**

Max VCC current before configuration (AUX output on): 0.225A  
 Max VCC current after configuration turns off AUX output: 0.165A  
 Max VDD current: 0.35mA

Key components and connections include:

- LMH6518** (U1008\_1) configured as a PGA.
- Power Supply:** +VPGA, +3V3\_PGA, +VBIAS.
- Resistors:** R1033\_1 (10K), R1045\_1 (1K), R1046\_1 (1K), R52\_1 (0), R97\_1 (0).
- Capacitors:** C1033\_1 (100nF), C1041\_1 (100nF), C1042\_1 (100nF), C1040\_1 (10uF), C1039\_1 (10uF), C1038\_1 (1uF), C1037\_1 (1uF), C1036\_1 (1uF), C1043\_1 (1uF).
- Trimmer:** FB1001\_1 (120@100MHZ).
- Output:** OUT\_P, OUT\_N, BUF\_OUT\_R.

# ADC Driver

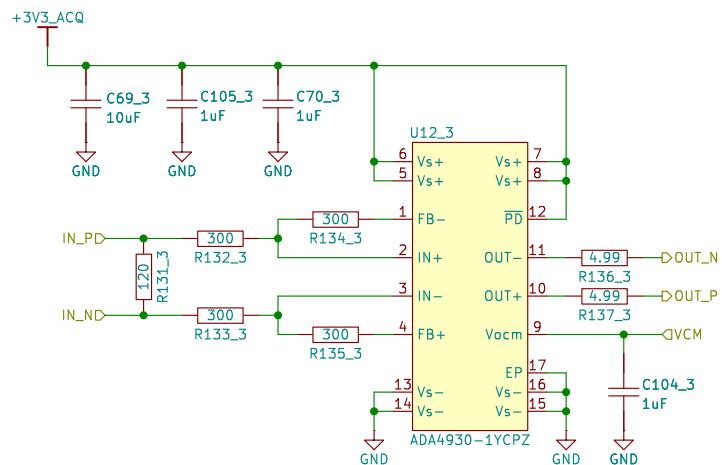


|                                                     |  |                            |  |
|-----------------------------------------------------|--|----------------------------|--|
| EEVengers                                           |  | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ADC Driver 4/<br>File: ADC_Driver.kicad_sch |  |                            |  |
| Title: ThunderScope                                 |  |                            |  |
| Size: A4                                            |  | Date:                      |  |
| KiCad E.D.A. 9.0.3                                  |  | Rev: 5.3<br>Id: 3/20       |  |

**ADC Driver**

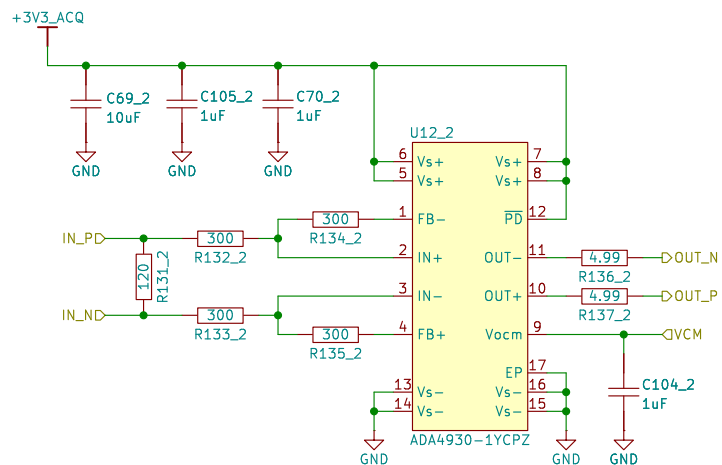
The schematic diagram illustrates an ADC Driver circuit centered around the ADA4930-1YCPZ operational amplifier (U12\_3). The circuit is powered by a +3V3\_ACQ supply, which is decoupled by three capacitors: C69\_3 (10uF), C105\_3 (1uF), and C70\_3 (1uF). The op-amp's non-inverting input (IN+) is connected to the IN\_PD input through a 300 ohm resistor (R132\_3). The inverting input (IN-) is connected to the IN\_ND input through a 300 ohm resistor (R133\_3). The feedback network consists of two 300 ohm resistors (R134\_3 and R135\_3) connecting the outputs (OUT- and OUT+) back to the inputs. The op-amp's output pins (OUT- and OUT+) are connected to the OUT\_N and OUT\_P outputs through 4.99 ohm resistors (R136\_3 and R137\_3). The common-mode voltage (Vcm) pin is connected to a common-mode voltage source (VCM). The op-amp's power supply pins (Vs+, Vs-, and EP) are connected to the +3V3\_ACQ supply and ground (GND). A decoupling capacitor C104\_3 (1uF) is connected to the EP pin. The op-amp is labeled ADA4930-1YCPZ.

|                                                     |       |                              |
|-----------------------------------------------------|-------|------------------------------|
| <b>EEVengers</b>                                    |       | Drawn by: Aleksa Bjelogrljic |
| Sheet: /ADC Driver 3/<br>File: ADC_Driver.kicad_sch |       |                              |
| <b>Title: ThunderScope</b>                          |       |                              |
| Size: A4                                            | Date: | <b>Rev: 5.3</b>              |
| KiCad E.D.A. 9.0.3                                  |       | Id: 3/20                     |



|                                                     |  |                            |                 |
|-----------------------------------------------------|--|----------------------------|-----------------|
| <b>EEVengers</b>                                    |  | Drawn by: Aleksa Bjelogrić |                 |
| Sheet: /ADC Driver 3/<br>File: ADC_Driver.kicad_sch |  |                            |                 |
| <b>Title: ThunderScope</b>                          |  |                            |                 |
| Size: A4                                            |  | Date:                      | Rev: <b>5.3</b> |
| KiCad E.D.A. 9.0.3                                  |  |                            | Id: 3/20        |

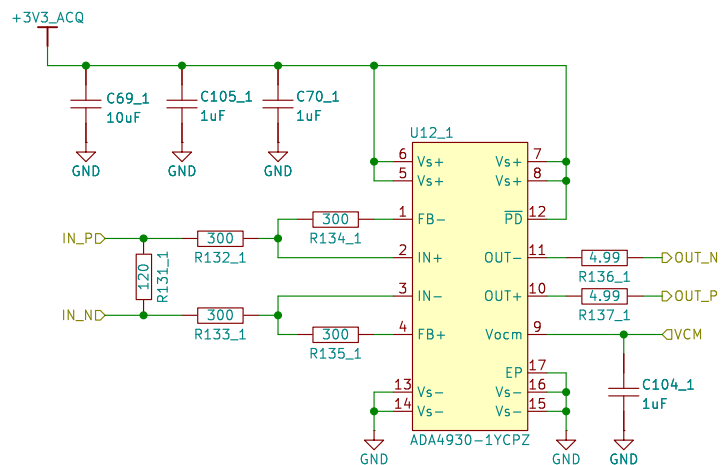
# ADC Driver



|                                                     |  |                            |  |
|-----------------------------------------------------|--|----------------------------|--|
| EEVengers                                           |  | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ADC Driver 2/<br>File: ADC_Driver.kicad_sch |  |                            |  |
| Title: <b>ThunderScope</b>                          |  |                            |  |
| Size: A4                                            |  | Date:                      |  |
| KiCad E.D.A. 9.0.3                                  |  | Rev: 5.3<br>Id: 3/20       |  |

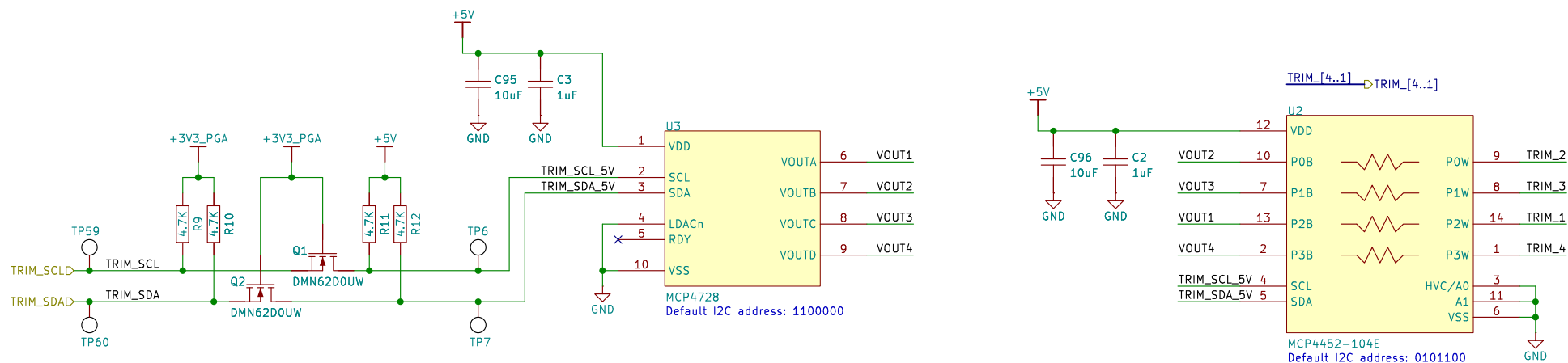


# ADC Driver

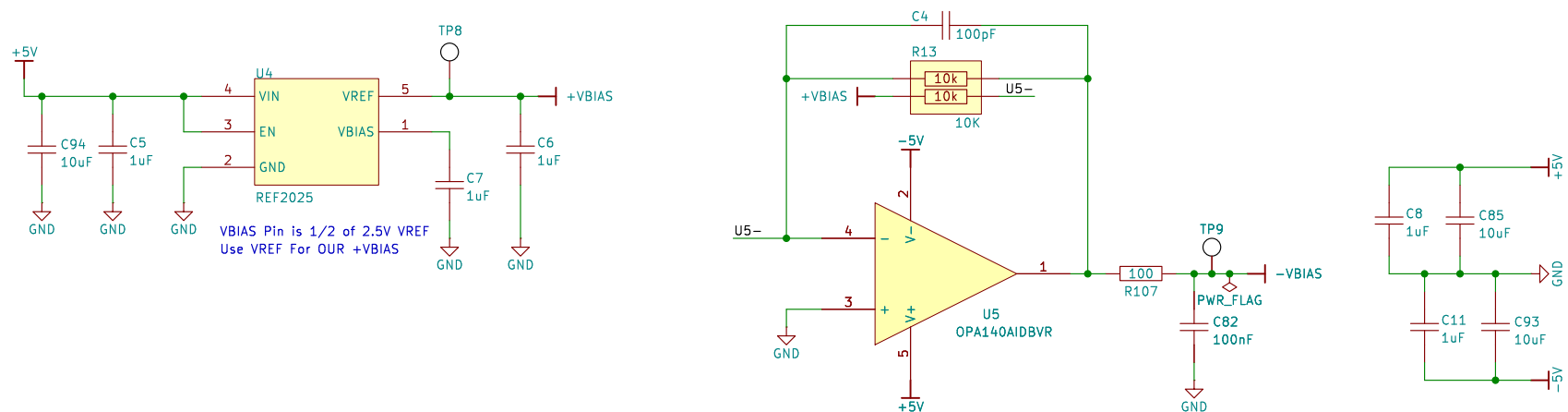


|                                                     |  |                            |  |
|-----------------------------------------------------|--|----------------------------|--|
| EEVengers                                           |  | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ADC Driver 1/<br>File: ADC_Driver.kicad_sch |  |                            |  |
| Title: ThunderScope                                 |  |                            |  |
| Size: A4                                            |  | Date:                      |  |
| KiCad E.D.A. 9.0.3                                  |  | Rev: 5.3<br>Id: 3/20       |  |

## Offset Voltage Trim and User Offset Control

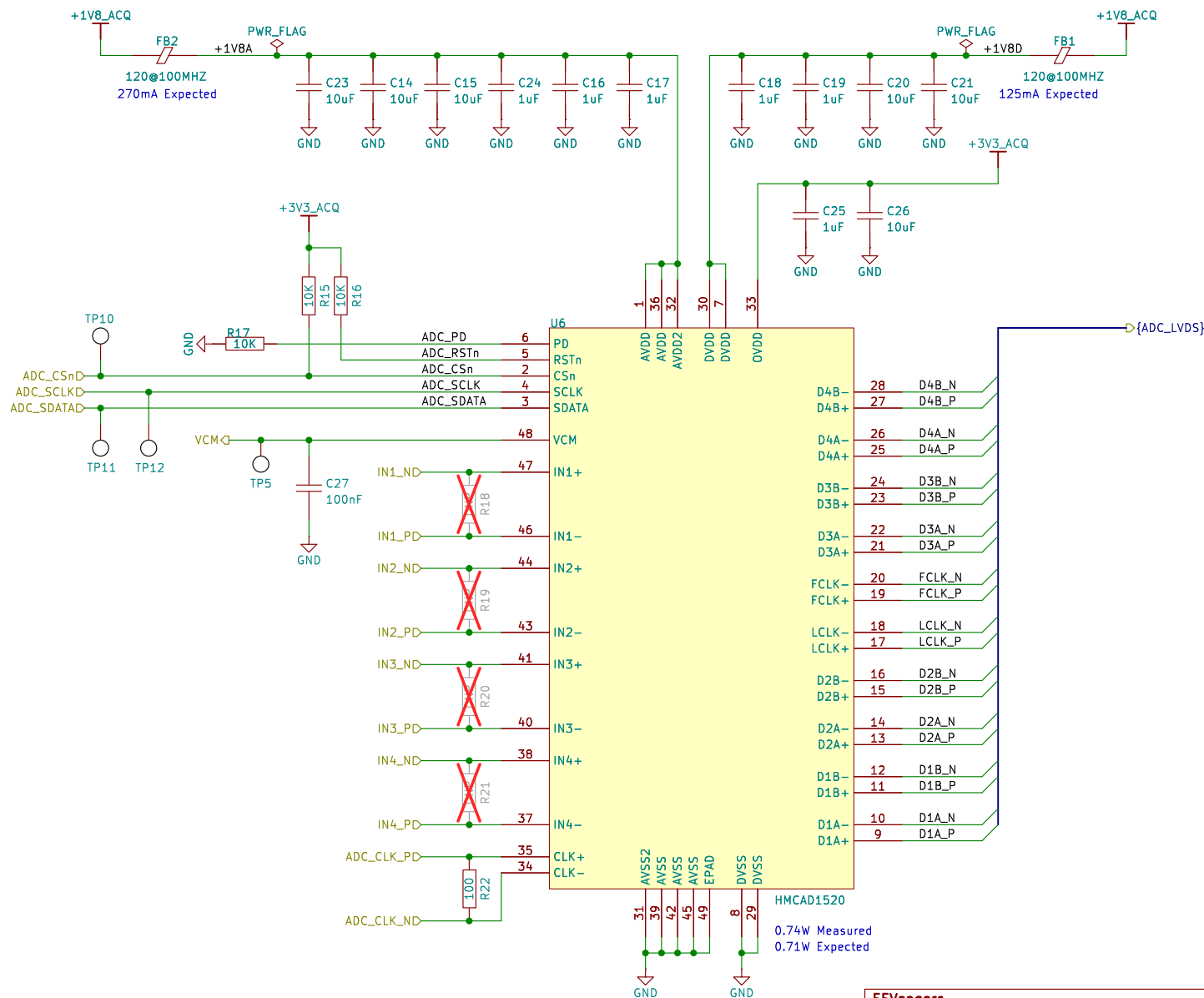


## Bias Voltage Generation



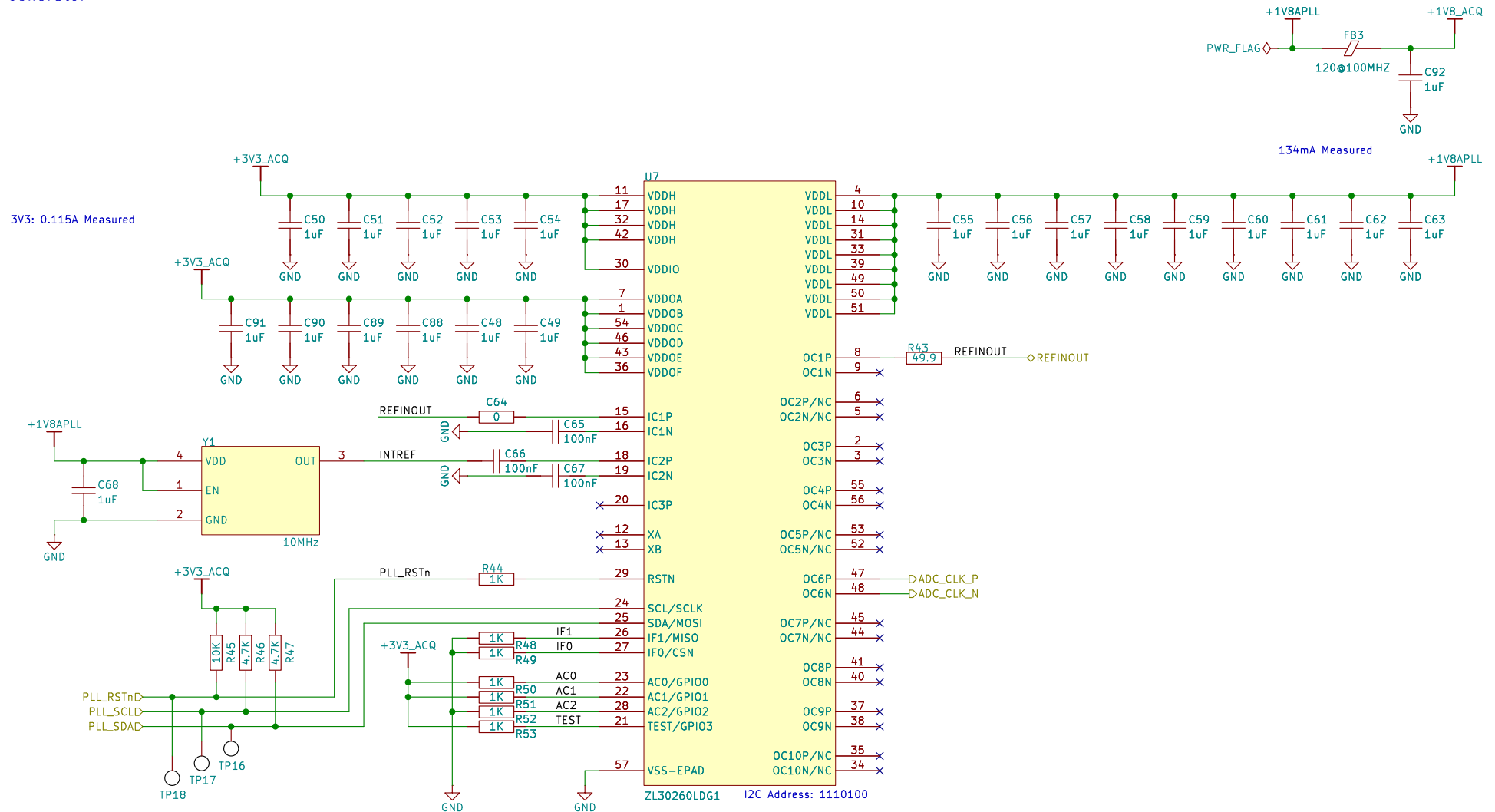
|                                                        |       |                            |          |
|--------------------------------------------------------|-------|----------------------------|----------|
| EEVengers                                              |       | Drawn by: Aleksa Bjelogrić |          |
| Sheet: /Front End Trim and Bias/<br>File: FE.kicad_sch |       |                            |          |
| Title: ThunderScope                                    |       |                            |          |
| Size: A4                                               | Date: |                            | Rev: 5.3 |
| KiCad E.D.A. 9.0.3                                     |       |                            | Id: 4/20 |

# ADC



|                     |       |                            |          |
|---------------------|-------|----------------------------|----------|
| EEVengers           |       | Drawn by: Aleksa Bjelogrić |          |
| Sheet: /ADC/        |       |                            |          |
| File: ADC.kicad_sch |       |                            |          |
| Title: ThunderScope |       |                            |          |
| Size: A4            | Date: |                            | Rev: 5.3 |
| KiCad E.D.A. 9.0.3  |       |                            | Id: 5/20 |

## Clock Generator



| IF1 | IF0 | Processor Interface                                 | Configuration Memory to Use |
|-----|-----|-----------------------------------------------------|-----------------------------|
| 0   | 0   | I <sup>2</sup> C, slave address 11101 00            | Internal ROM                |
| 0   | 1   | I <sup>2</sup> C, slave address 11101 01            | Internal ROM                |
| 1   | 0   | SPI Slave                                           | Internal ROM                |
| 1   | 1   | SPI Master during auto-configuration then SPI Slave | External SPI EEPROM         |

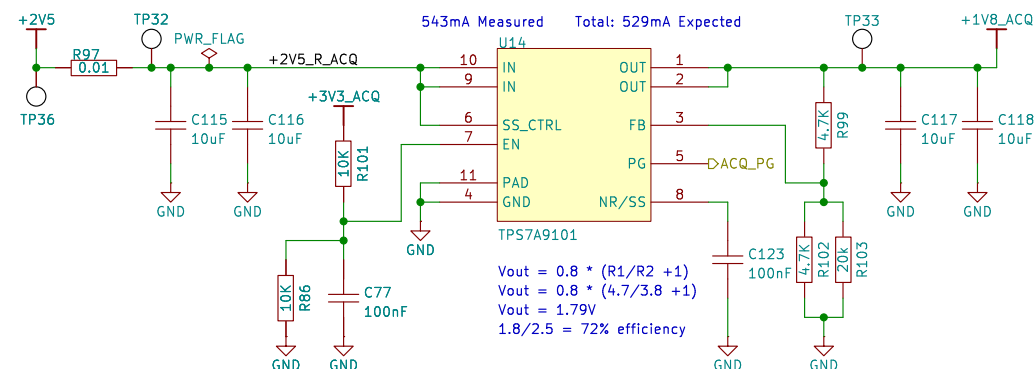
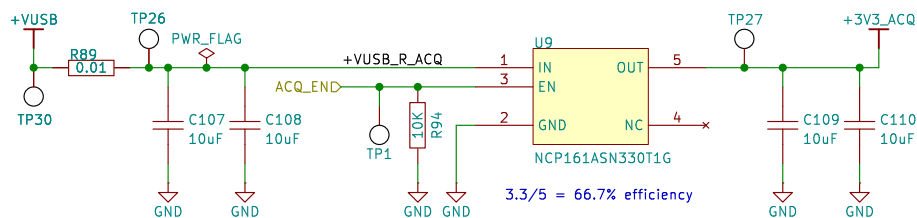
To configure the device as specified in the first three rows above but *without* auto-configuring from internal ROM, wire devices pins as follows: TEST=1 and AC[2:0]=011, as described in section 5.2.

| AC2 | AC1 | AC0 | Auto Configuration |
|-----|-----|-----|--------------------|
| 0   | 0   | 0   | Configuration 0    |
| 0   | 0   | 1   | Configuration 1    |
| 0   | 1   | 0   | Configuration 2    |
| 0   | 1   | 1   | Configuration 3    |
| 1   | 0   | 0   | Configuration 4    |
| 1   | 0   | 1   | Configuration 5    |
| 1   | 1   | 0   | Configuration 6    |
| 1   | 1   | 1   | Configuration 7    |

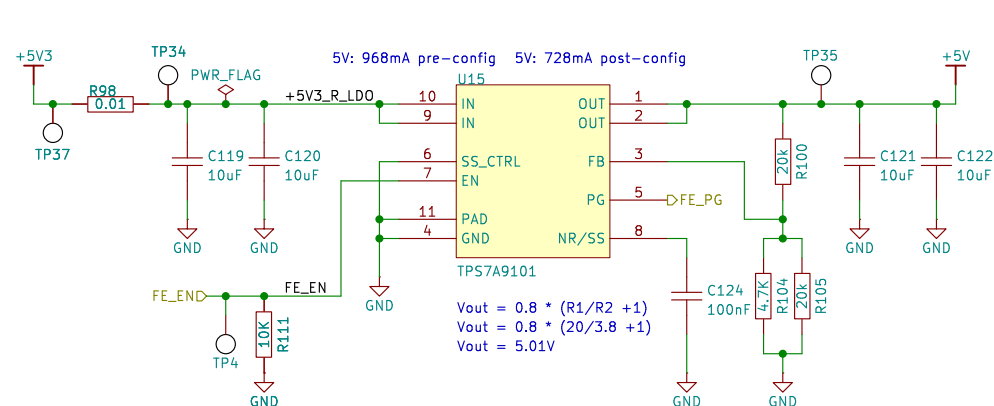
3.3 + 1.8V operation w/ one input one output: 0.67W Expected  
0.62W Measured

|                                                 |       |                            |  |
|-------------------------------------------------|-------|----------------------------|--|
| <b>EEVengers</b>                                |       | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /Clock Generator/<br>File: PLL.kicad_sch |       |                            |  |
| <b>Title: ThunderScope</b>                      |       |                            |  |
| Size: A4                                        | Date: | Rev: 5.3                   |  |
| KiCad E.D.A. 9.0.3                              |       | Id: 6/20                   |  |

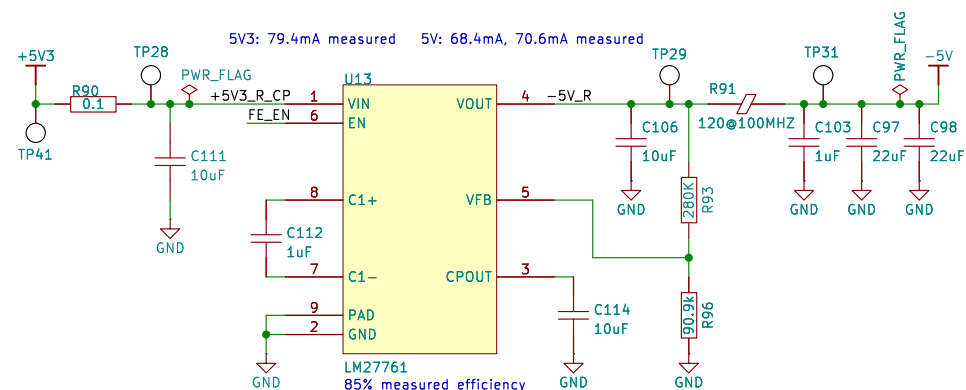
## Acquisition Voltage Regulators



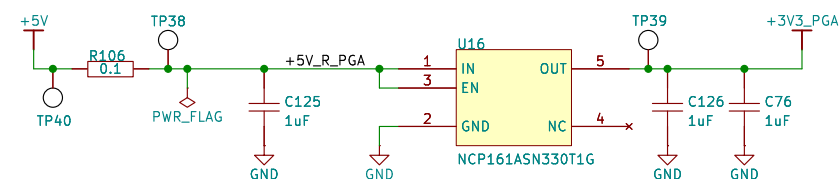
## Front End Voltage Regulators



6ms soft start time on previous LDO, need same or greater

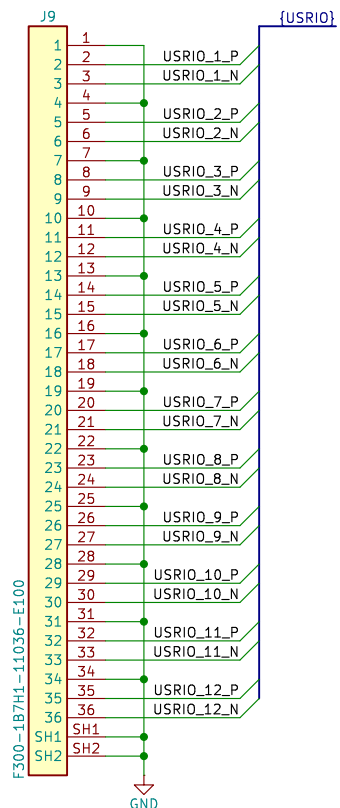
$$T_{SS} = (V_{REF} \times C_{nr/ss}) / I_{nr/ss}$$
$$T_{SS} = (0.8 \times 100nF) / 6.2\mu A \text{ [for SS_CTRL = GND]}$$
$$T_{SS} = 0.0129s = 13ms$$


$V_{out} = -1.22V \cdot (R_1 + R_2) / R_2$   
 $V_{out} = -1.22 \cdot (280 / 90.9 + 1) = -4.98V$   
 The value for  $R_2$  must be no less than 50 kΩ.



|                                                                 |       |                            |  |
|-----------------------------------------------------------------|-------|----------------------------|--|
| <b>EEVengers</b>                                                |       | Drawn by: Aleksa Bjelogrić |  |
| Sheet: /ACQ and FE Voltage Regs/<br>File: ACQ_FE_VREG.kicad_sch |       |                            |  |
| <b>Title: ThunderScope</b>                                      |       |                            |  |
| Size: A4                                                        | Date: | Rev: <b>5.3</b>            |  |
| KiCad E.D.A. 9.0.3                                              |       | Id: 7/20                   |  |

## User IO Flat Flex Connector



### FPGA Power Inputs



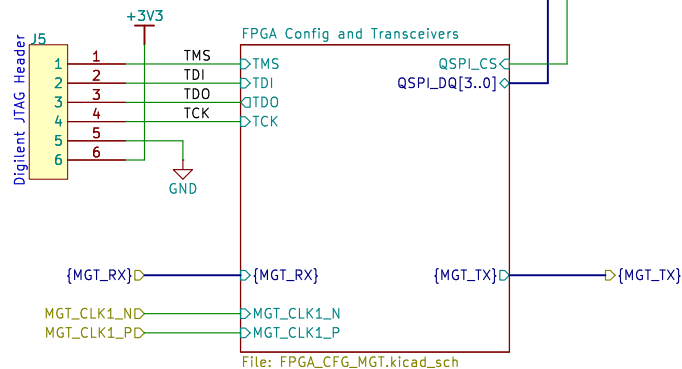
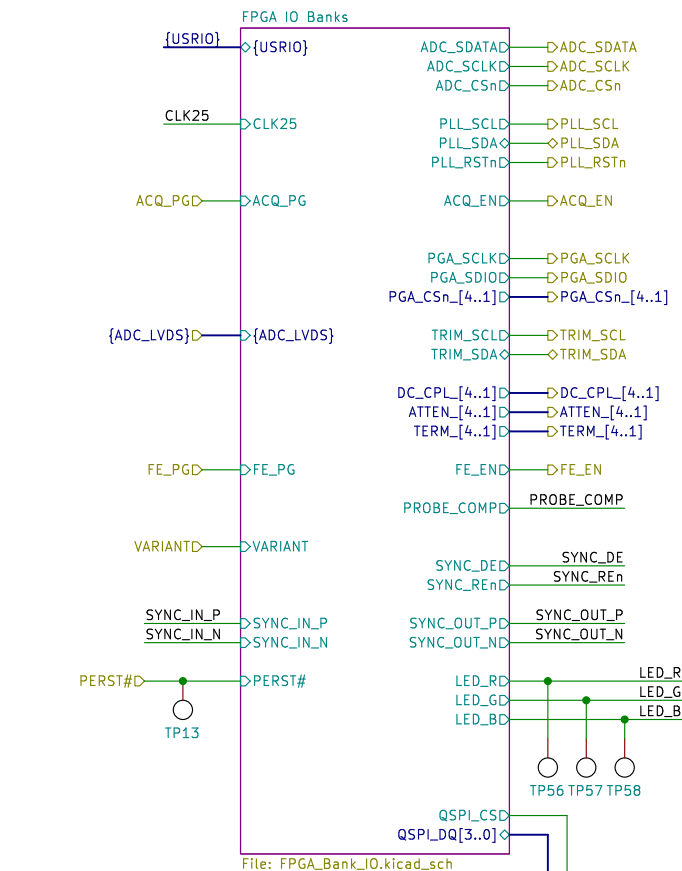
File: FPGA\_PWR.kicad\_sch

### FPGA Voltage Regs

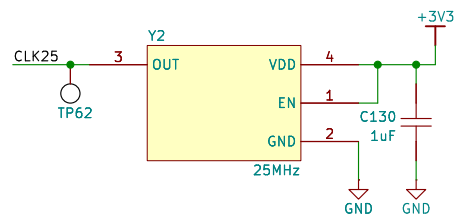


File: FPGA\_VREG.kicad\_sch

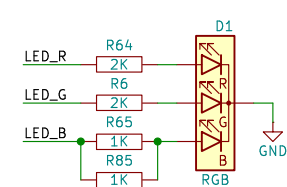
## FPGA



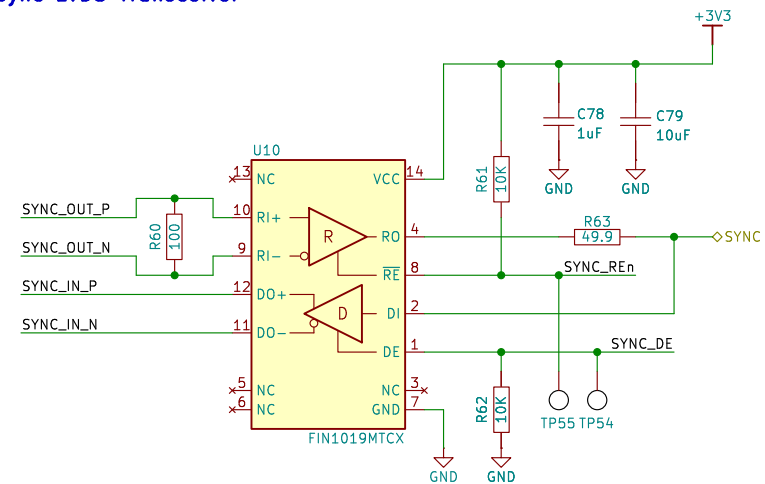
## Always On Clock



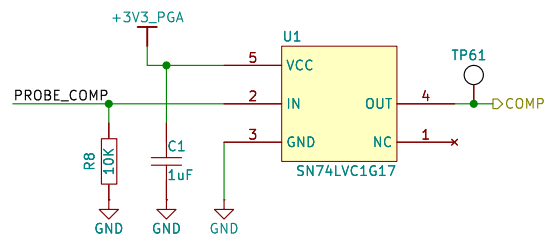
## Status LED



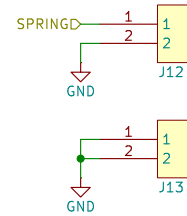
## Sync LVDS Transceiver



## Probe Compensation



## Spring Connectors



EEVengers

Drawn by: Aleksa Bjelogrić

Sheet: /FPGA/

File: FPGA.kicad\_sch

**Title: ThunderScope**

Size: A4

Date:

KiCad E.D.A. 9.0.3

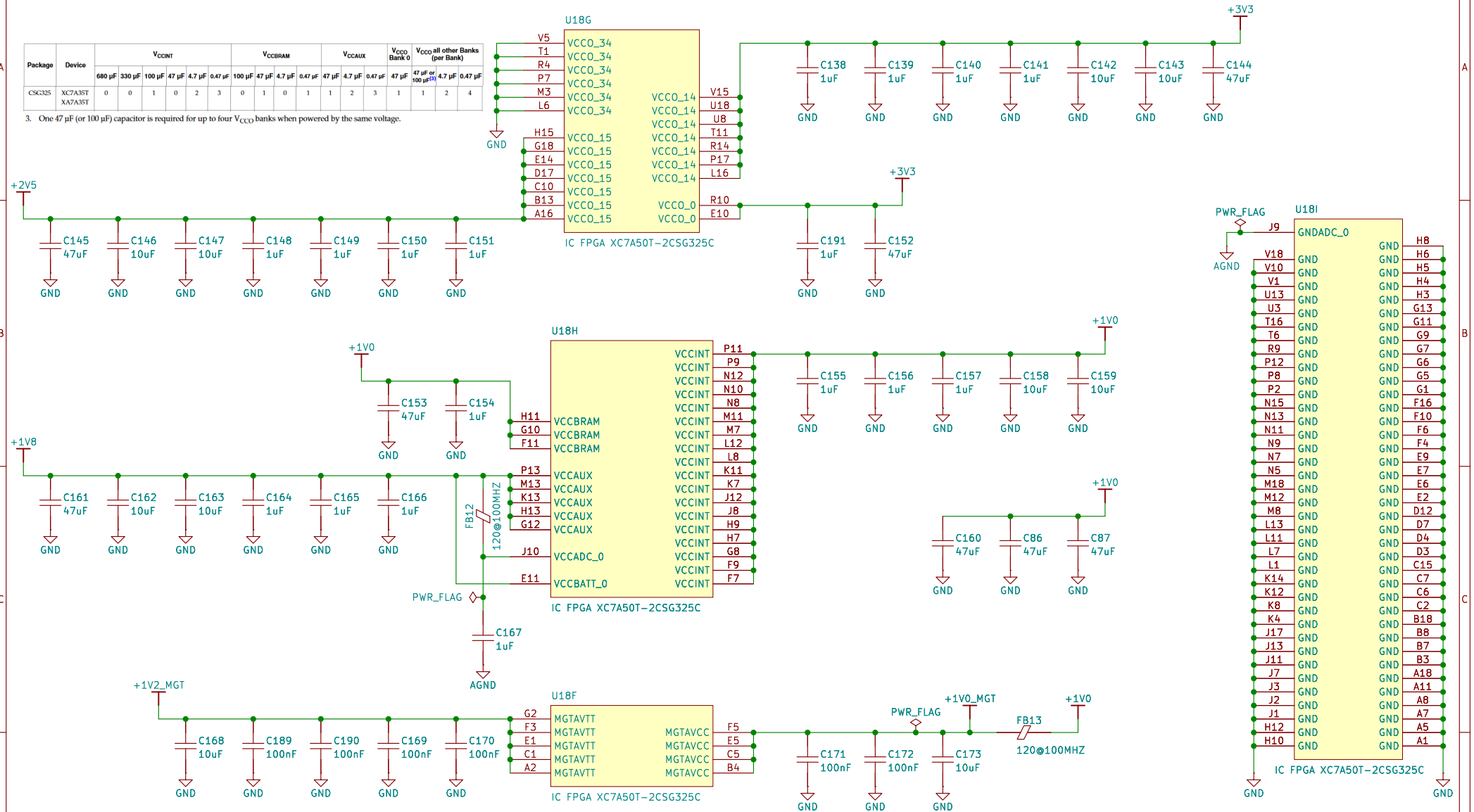
Rev: 5.3

Id: 8/20

## FPGA Power Inputs

| Package | Device             | V <sub>CCINT</sub> |        |        |       |        |         | V <sub>CCBRAM</sub> |       |        |         | V <sub>CCAUX</sub> |        |         | V <sub>CC0</sub><br>Bank 0 | V <sub>CC0</sub> all other Banks<br>(per Bank) |        |         |  |
|---------|--------------------|--------------------|--------|--------|-------|--------|---------|---------------------|-------|--------|---------|--------------------|--------|---------|----------------------------|------------------------------------------------|--------|---------|--|
|         |                    | 680 μF             | 330 μF | 100 μF | 47 μF | 4.7 μF | 0.47 μF | 100 μF              | 47 μF | 4.7 μF | 0.47 μF | 47 μF              | 4.7 μF | 0.47 μF | 47 μF                      | 47 μF<br>or<br>100 μF                          | 4.7 μF | 0.47 μF |  |
| CSG325  | XC7A35T<br>XA7A35T | 0                  | 0      | 1      | 0     | 2      | 3       | 0                   | 1     | 0      | 1       | 1                  | 2      | 3       | 1                          | 1                                              | 2      | 4       |  |

- One 47  $\mu\text{F}$  (or 100  $\mu\text{F}$ ) capacitor is required for up to four  $V_{\text{CCO}}$  banks when powered by the same voltage.



| Capacitor | Package Pins |         |     | Value       |
|-----------|--------------|---------|-----|-------------|
|           | MGTAVCC      | MGTAVTT | GND |             |
| Cap1      |              | F3      | F4  | 0.1 $\mu$ F |
| Cap2      |              | A2      | A1  |             |
| Cap3      | B4           |         | B3  |             |
| Cap4      | F5           |         | F6  |             |

| Qty/Power Supply Group |          | Capacitance (µF) | Tolerance | Type    |
|------------------------|----------|------------------|-----------|---------|
| MGTA1VCC               | MGTA1VTT |                  |           |         |
| 1                      | 1        | 4.7              | 10%       | Ceramic |
| 2                      | 2        | 0.1              | 10%       | Ceramic |

EEVengers

Drawn by: Aleksa Bjelogrić

Sheet: /FPGA/FPGA Power Inputs/  
File: FPGA\_PWR.kicad\_sch

**Title:** ThunderScope

Size: A4

Date:

KiCad E.D.A. 9.0.3

Rev: 5.3

Id: 11/20

## FPGA Voltage Regulators

The recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO  
1V,1V,1.8V,2.5V and 3.3V

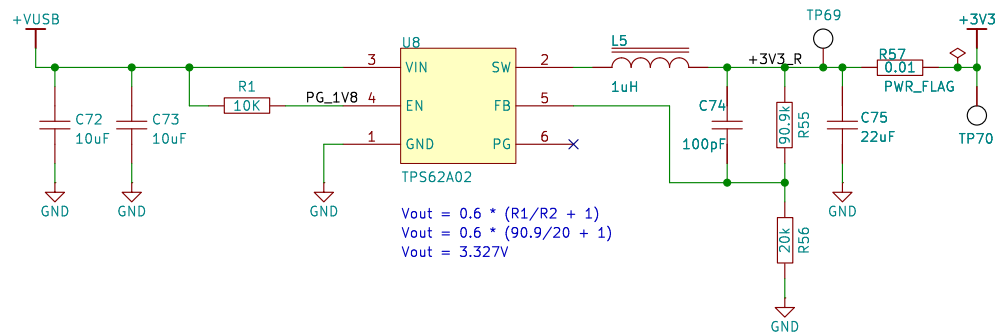
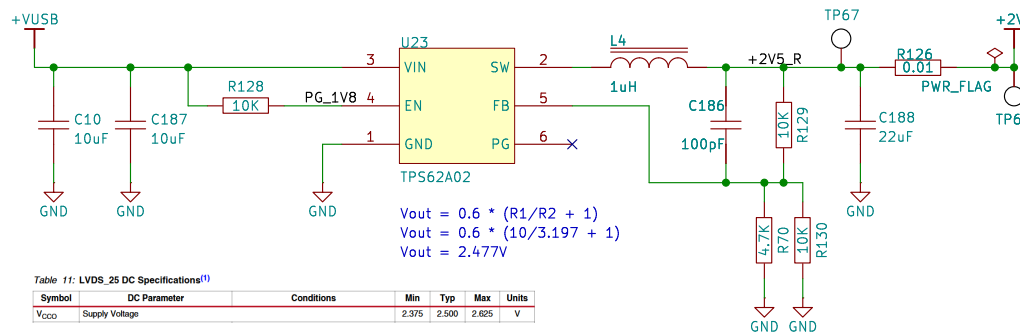
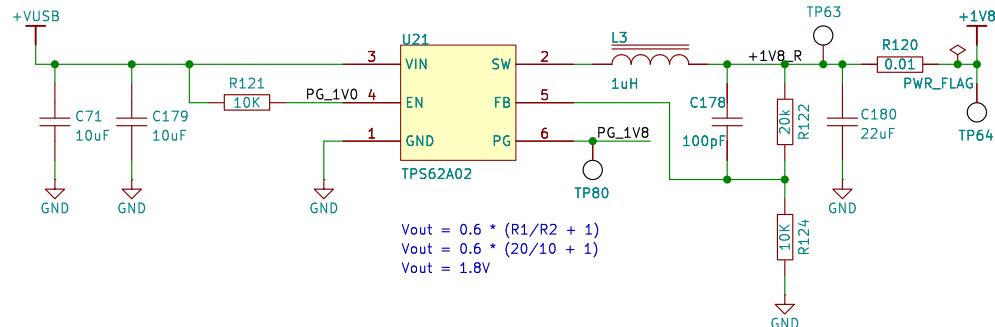
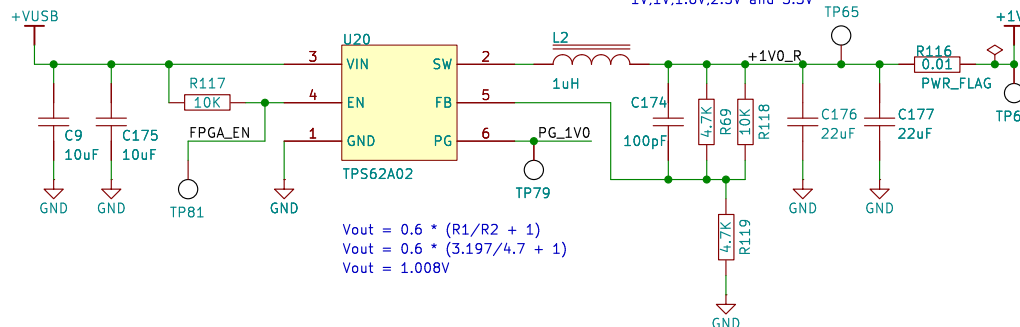
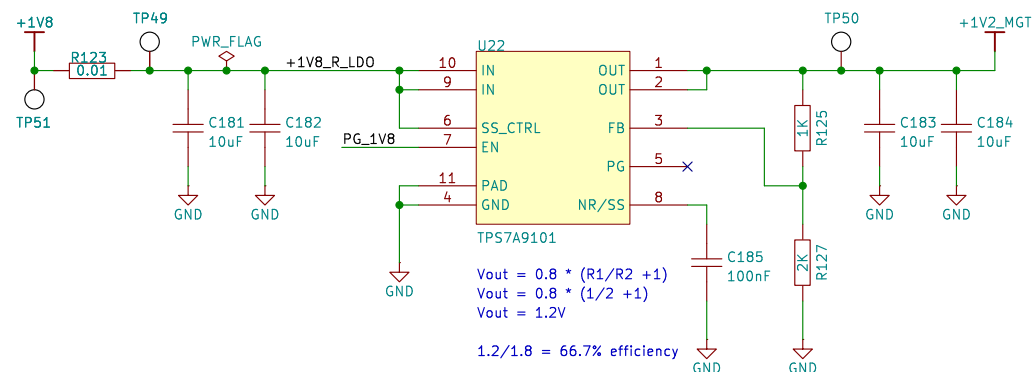


Table 11: LVDS\_25 DC Specifications<sup>(1)</sup>

| Symbol           | DC Parameter   | Conditions | Min   | Typ   | Max   | Units |
|------------------|----------------|------------|-------|-------|-------|-------|
| V <sub>CCO</sub> | Supply Voltage |            | 2.375 | 2.500 | 2.625 | V     |



The recommended power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VMGTAVTT  
1V,1V,1.2V

EEVengers

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Sheet: /FPGA/FPGA Voltage Regs/  
File: FPGA\_VREG.kicad\_sch

**Title: ThunderScope**

Size: A4

Date:

Rev: 5.3

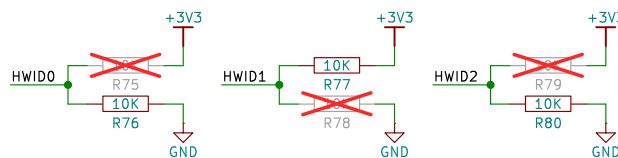
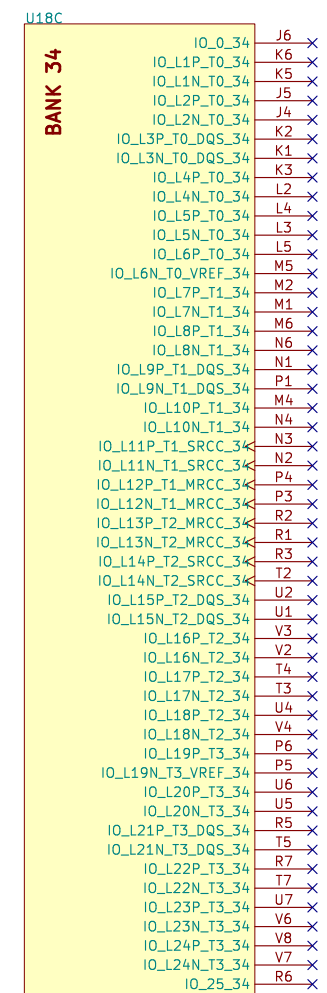
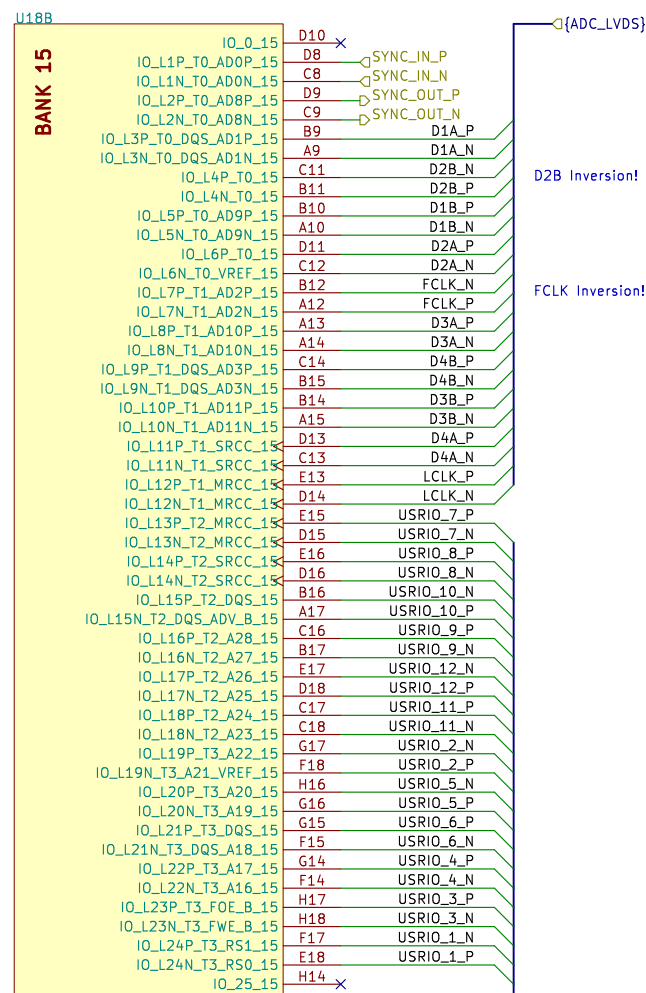
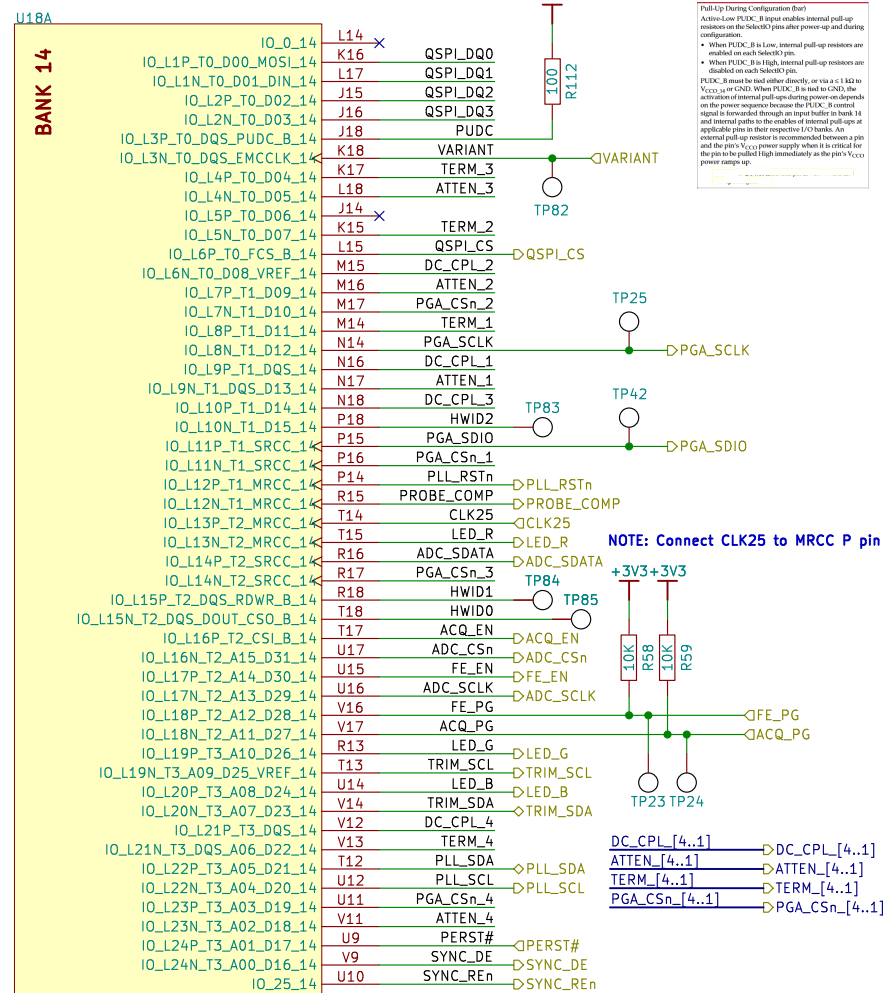
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Id: 12/20



# FPGA IO Banks

Full TP Coverage on Bank 14 Signals (except PUDC)



|                                                             |       |                            |                 |
|-------------------------------------------------------------|-------|----------------------------|-----------------|
| EEVengers                                                   |       | Drawn by: Aleksa Bjelogrić |                 |
| Sheet: /FPGA/FPGA IO Banks/<br>File: FPGA_Bank_IO.kicad_sch |       |                            |                 |
| Title: <b>ThunderScope</b>                                  |       |                            |                 |
| Size: A4                                                    | Date: |                            | Rev: <b>5.3</b> |
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FPGA Configuration

A High signal on the DONE pin indicates completion of the configuration sequence. The DONE output is an open-drain output by default.

**Note:** DONE has an internal pull-up resistor of approximately 10 kΩ. There is no setup/hold requirement for the DONE register. These changes, along with the DonePipe register software default, eliminate the need for the DriveDONE driver-option. External 330Ω resistor circuits are not required but can be used as they have been in previous generations.

Connect INIT\_B to a ≤ 4.7 kΩ pull-up resistor to V<sub>CC0\_0</sub> to ensure clean Low-to-High transitions.

Connect PROGRAM\_B to an external ≤ 4.7 kΩ pull-up resistor to V<sub>CC0\_0</sub> to ensure a stable High input, and

Table 2-1: 7 Series FPGA Configuration Modes

| Configuration Mode | M[2:0] | Bus Width  | CCLK Direction |
|--------------------|--------|------------|----------------|
| Master SPI         | 001    | x1, x2, x4 | Output         |

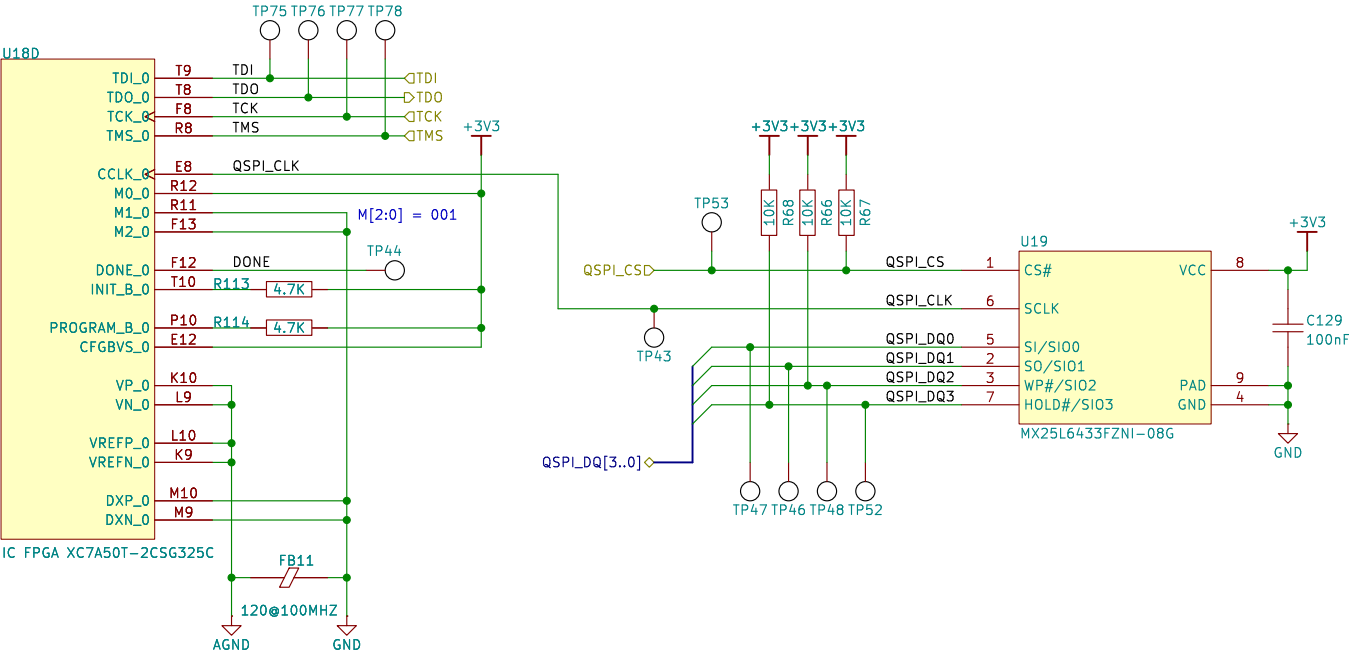
Table 2-6: Spartan-7, Artix-7 and Kintex-7 FPGA Configuration Mode, Compatible Voltages, and CFGBV5 Connection

| Configuration Mode        | Banks Used | Configuration Interface I/O Voltage | HB Bank 0 V <sub>CC0_0</sub> | HB Bank 14 V <sub>CC0_14</sub> | HB Bank 15 V <sub>CC0_15</sub> | CFGBV5             |
|---------------------------|------------|-------------------------------------|------------------------------|--------------------------------|--------------------------------|--------------------|
| JTAG (only)               | 0          | 3.3V                                | 3.3V                         | Any                            | Any                            | V <sub>CC0_0</sub> |
|                           |            | 2.5V                                | 2.5V                         | Any                            | Any                            | V <sub>CC0_0</sub> |
|                           |            | 1.8V                                | 1.8V                         | Any                            | Any                            | GND                |
|                           |            | 1.5V                                | 1.5V                         | Any                            | Any                            | GND                |
|                           |            | 3.3V                                | 3.3V                         | 3.3V                           | Any                            | V <sub>CC0_0</sub> |
| Serial, SPI, or SelectMAP | 0, 14†1    | 2.5V                                | 2.5V                         | 2.5V                           | Any                            | V <sub>CC0_0</sub> |
|                           |            | 1.8V                                | 1.8V                         | 1.8V                           | Any                            | GND                |
|                           |            | 1.5V                                | 1.5V                         | 1.5V                           | Any                            | GND                |
|                           |            | 3.3V                                | 3.3V                         | 3.3V                           | 3.3V                           | V <sub>CC0_0</sub> |
|                           |            | 2.5V                                | 2.5V                         | 2.5V                           | 2.5V                           | V <sub>CC0_0</sub> |
| BPI†2                     | 0, 14, 15  | 1.8V                                | 1.8V                         | 1.8V                           | 1.8V                           | GND                |
|                           |            | 1.5V                                | 1.5V                         | 1.5V                           | 1.5V                           | GND                |

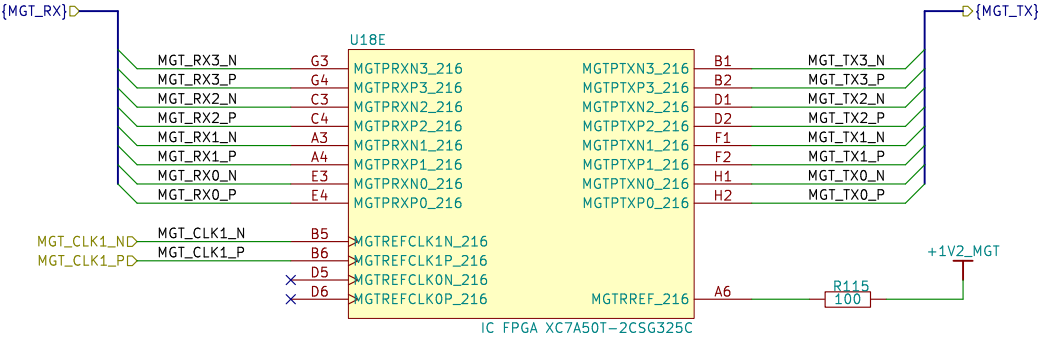
Notes:

1. BPI†2 for Multiboot or Fallback are in bank 15 but are typically only used in BPI mode and not supported in SPI mode.

2. BPI mode is not available in the Spartan-7 family.



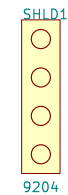
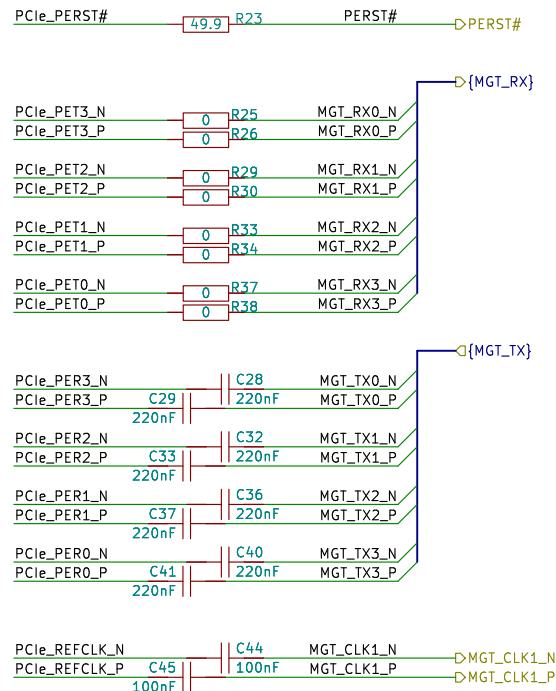
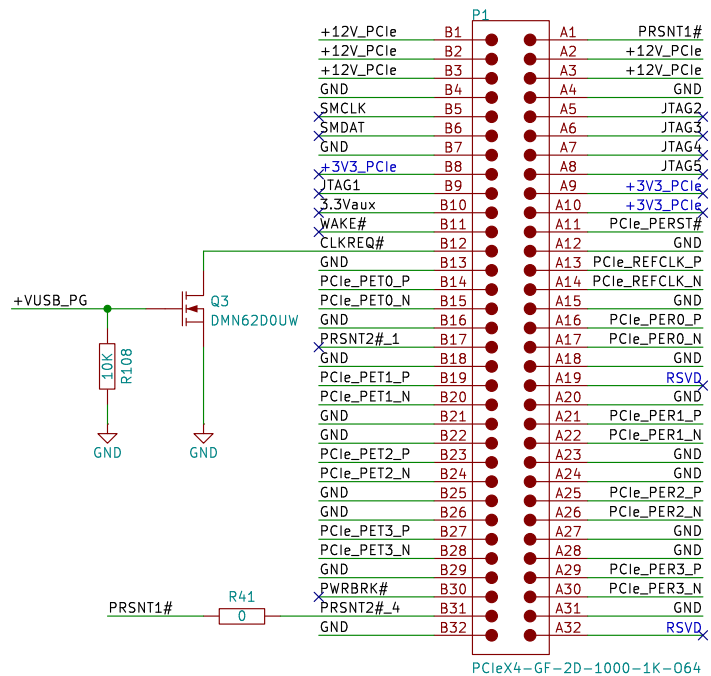
FPGA Transceivers



## PCIe x4 Edge Connector

AC Couple PER Lines with 220nF  
AC Couple REFCLK Lines with 100nF

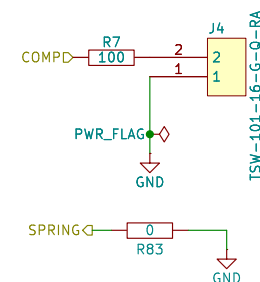
## PCIe bracket



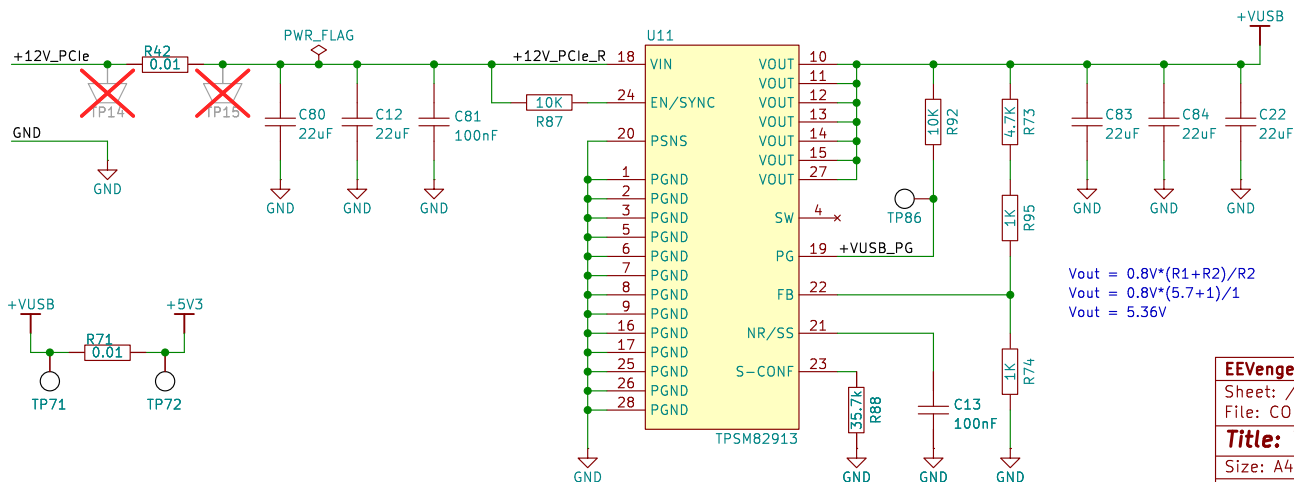
## Variant Pin Strap



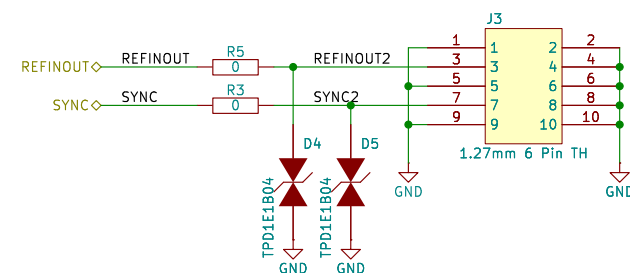
## Probe Comp



## 5V3 Buck Converter



## Refclock and Sync Header



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Sheet: /TS-PCIe Components/  
File: CON\_PCl\_e\_X4.kicad\_sch

Title: ThunderScope

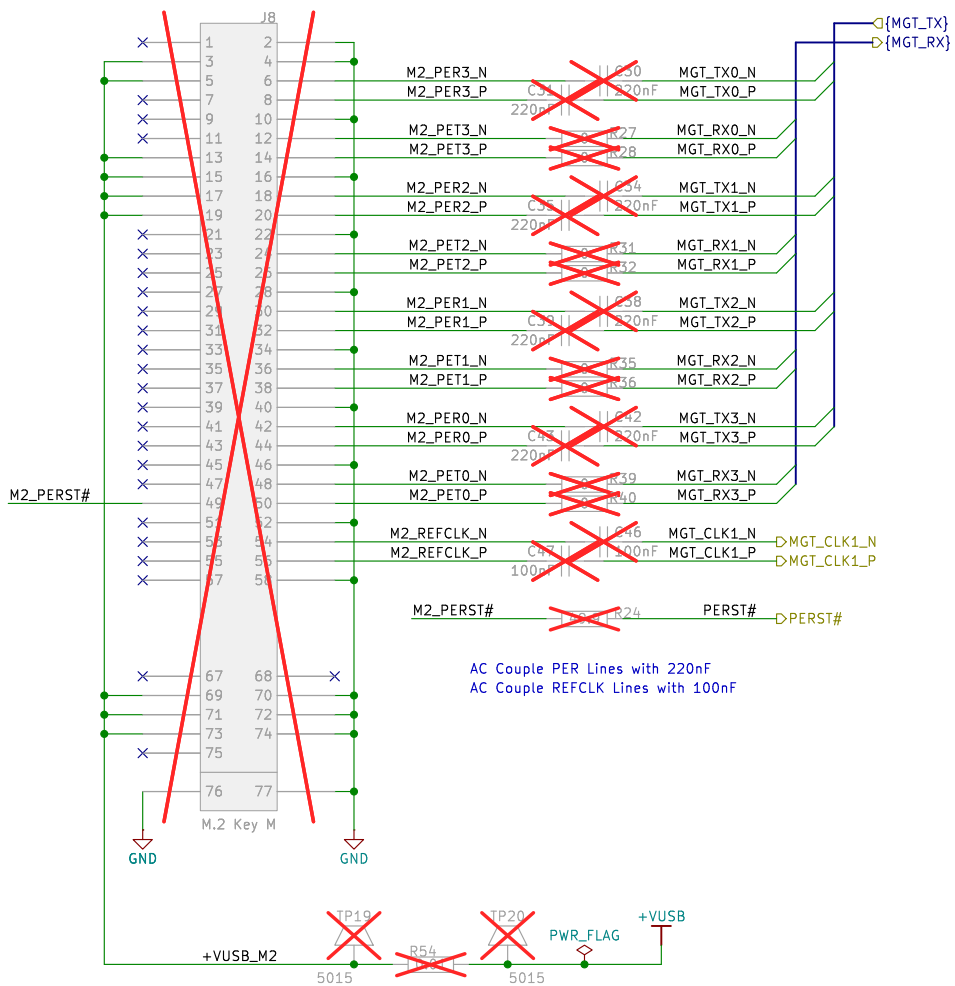
Size: A4 Date:

KiCad E.D.A. 9.0.3

Rev: 5.3

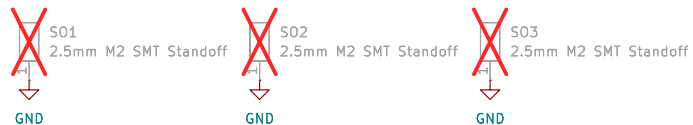
Id: 13/20

## M.2 Key M Connector – Custom Pinout



NOTE: The TB/USB4 adaptor must be modified to give us VUSB instead of 3V3

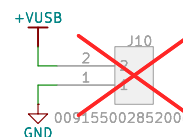
## Interposer Standoffs



## Probe Comp



## Fan Connector



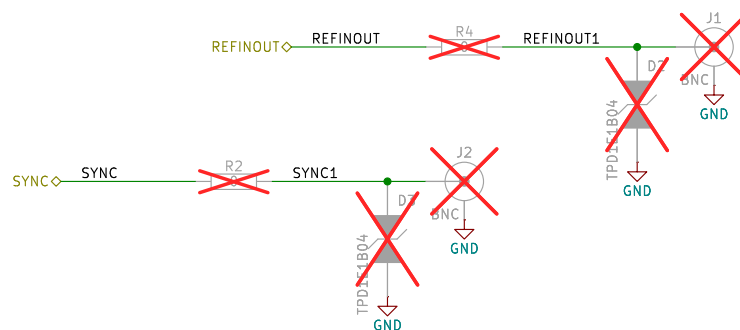
## Ground Lug



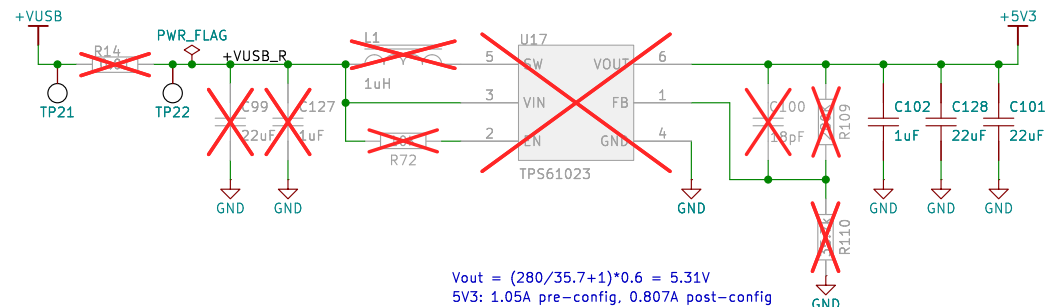
## Variant Pin Strap



## Refclock and Sync BNCs



## 5V3 Boost Converter



|                                                         |       |                            |           |
|---------------------------------------------------------|-------|----------------------------|-----------|
| EEVengers                                               |       | Drawn by: Aleksa Bjelogrić |           |
| Sheet: /TS—USB4 Components/<br>File: M2_KEY_M.kicad_sch |       |                            |           |
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| Size: A4                                                | Date: |                            | Rev: 5.3  |
| KiCad E.D.A. 9.0.3                                      |       |                            | Id: 14/20 |